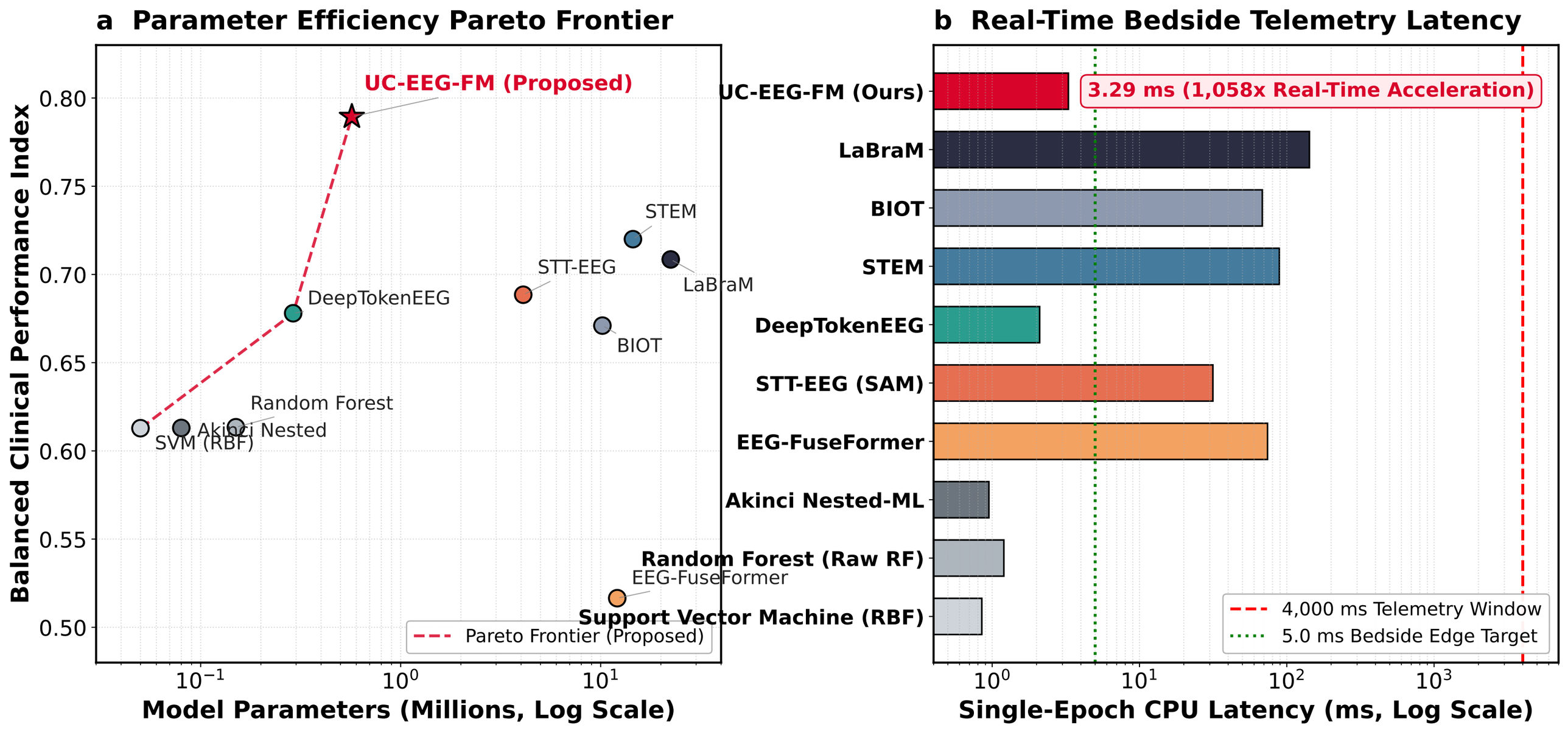
a Parameter Efficiency Pareto Frontier
UC-EEG-FM (Proposed)
DeepTokenEEG
STEM
STT-EEG
LaBraM
BIOT
Random Forest
Akinci Nested
SVM (RBF)
EEG-FuseFormer
Pareto Frontier (Proposed)
Balanced Clinical Performance Index
0.80
0.75
0.70
0.65
0.60
0.55
0.50
10^{-1}
10^{0}
10^{1}
Model Parameters (Millions, Log Scale)
b Real-Time Bedside Telemetry Latency
UC-EEG-FM (Ours)
3.29 ms (1,058x Real-Time Acceleration)
LaBraM
BIOT
STEM
DeepTokenEEG
STT-EEG (SAM)
EEG-FuseFormer
Akinci Nested-ML
Random Forest (Raw RF)
Support Vector Machine (RBF)
4,000 ms Telemetry Window
5.0 ms Bedside Edge Target
10^{0}
10^{1}
10^{2}
10^{3}
Single-Epoch CPU Latency (ms, Log Scale)